EXPERIMENTS IN MODERN PHYSICS

Second Edition

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ACADEMIC PRESS

An imprint of Elsevier Science

Amsterdam Boston London New York Oxford Paris San Diego
San Francisco Singapore Sydney Tokyo

Very different patterns are formed near and far from the aperture. In the near field we speak of *Fresnel diffraction*, and to observe the pattern it is convenient to form an *image* of it on a screen. In the far field we obtain the *Fraunhofer diffraction* pattern, which can be observed by simply placing a screen at some distance from the aperture; more precisely a lens should be used and the pattern observed in the focal plane. In the following three sections we discuss Fraunhofer diffraction from a slit and a circular aperture. The results shown were obtained with a CCD camera.

The diffraction grating was already introduced in Chapter 1. In Section 5.5 we derive the grating equation and show a modern setup that can be readily digitized; also included are results on the Hg spectrum. Next we introduce the concept of "spatial frequency" components in a beam of light. This allows us to manipulate an image by imposing suitable spatial filters, in the focal plane, a procedure also referred to as "Fourier optics." We have kept the mathematics simple and emphasized the physical principles instead. In Section 5.7 we discuss the Faraday effect, namely the rotation of the linear polarization of light when traversing a medium immersed in an axial magnetic field. The power of the lock-in detection technique is evident in this experiment. The last section is a demonstration and measurement of "Berry's phase." This is the rotation of polarization due to a topological change in the direction of propagation of the light. It is demonstrated by injecting the light in an optical fiber that is wound as a helix.

5.2. DIFFRACTION FROM A SLIT

We can find the minima of the diffraction pattern with the help of the sketches shown in Fig. 5.1. Consider a plane wave of visible light incident on a vertical slit of width d. The incident "rays" are normal to the screen that contains the slit; i.e., the *wavefront* is parallel to the screen. We can "divide" the slit in half (i.e., $AB = BC = d/2$) and consider the rays 1 and 2 emerging at an angle θ with respect to the direction of incidence (Fig. 5.1a). The path difference between these rays is $BD = AB \sin\theta = (d/2)\sin\theta$. If the path difference is $\lambda/2$, then at *large distances* from the slit, rays 1 and 2 will interfere destructively. However, this will also happen for rays $1'$ and $2'$, $1''$ and $2''$, and so on, so that there will be no light in the direction θ_1, where

$$\frac{d}{2}\sin\theta_1 = \frac{\lambda}{2}. \tag{5.1}$$

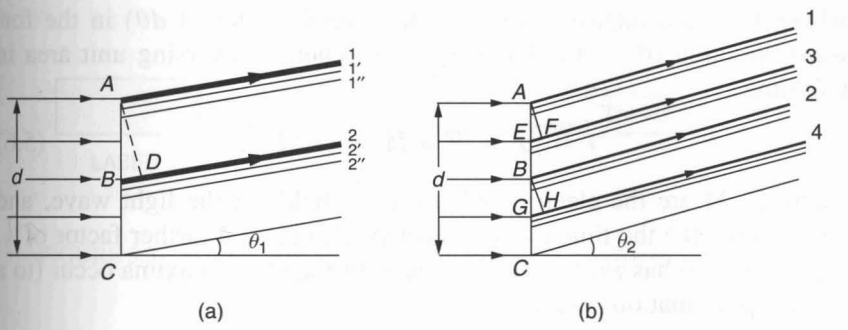


FIGURE 5.1 Finding the minima of a diffraction pattern (a) the slit of width d is "divided" in half and (b) into quarters. The rays are focused at infinity and the path difference is indicated.

In contrast, at $\theta = 0$ the path length (out to a large distance) of all rays is equal and the resultant amplitude is maximal.

To find the next zero, let us "divide" the slit into quarters as shown in Fig. 5.1b. In this case ray 1 will interfere destructively with ray 3 when $EF = AE \sin\theta = (d/4)\sin\theta = \lambda/2$. However, ray 2 also will interfere destructively with ray 4 and similarly for all intermediate rays. Thus there will be no light in the direction θ_2, where

$$\frac{d}{4}\sin\theta_2 = \frac{\lambda}{2}. \tag{5.2}$$

This argument can be continued by subdividing the slit into an (integral) number of smaller and smaller segments. By analogy with Eqs. (5.1) and (5.2) we find the generalized expression for the minima

$$d\sin\theta_n = \pm n\lambda, \qquad n = 1, 2, 3, \ldots. \tag{5.3}$$

For small angles $\sin\theta \sim \theta$ and

$$\theta_n = n\frac{\lambda}{d}, \qquad n = 1, 2, 3, \ldots. \tag{5.4}$$

The complete expression for the intensity distribution of the diffracted light is derived in the next section. It is

$$I(\theta) = I_0\left[\frac{\sin\left(\frac{\pi d}{\lambda}\sin\theta\right)}{\frac{\pi}{\lambda}\sin\theta}\right]^2, \tag{5.5}$$

where I_0 is the intensity (into a small angular interval $d\theta$) in the forward direction ($\theta = 0$). Intensity is the energy traversing unit area in unit time

$$I = |\mathbf{S}| = |\mathbf{E} \times \mathbf{H}| = c\epsilon_0 |E|^2, \tag{5.6}$$

where $\mathbf{E}$, $\mathbf{H}$ are the electric and magnetic fields of the light wave, and we usually take the time average, which introduces a further factor of $\frac{1}{2}$. Equation (5.5) has zeros in agreement with Eq. (5.3); maxima occur (to a good approximation) when

$$d \sin\theta_m = \pm\left(m + \frac{1}{2}\right)\lambda, \qquad m = 1, 2, 3, \ldots. \tag{5.7}$$

The intensity at the secondary maxima decreases as m increases. Equation (5.5) is of the general form

$$f(x) = \frac{\sin^2 x}{x^2}, \tag{5.8}$$

which is graphed in Fig. 5.2. Note that as $x \to 0$, $f(x) \to 1$.

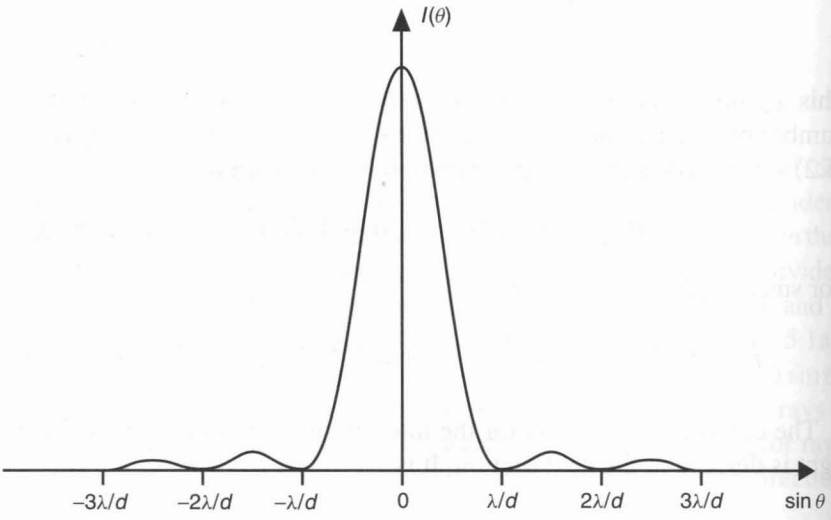


FIGURE 5.2 Plot of the Fraunhofer diffraction pattern $\sin^2 x/x^2$.

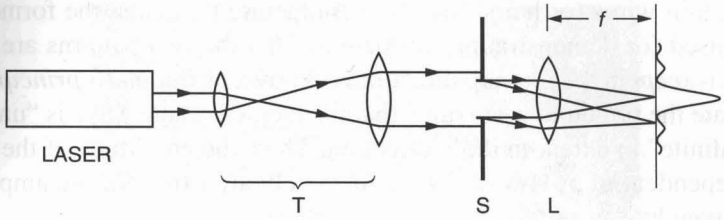


FIGURE 5.3 Schematic of a simple layout to observe Fraunhofer diffraction.

The experimental setup is shown in Fig. 5.3. The laser beam is expanded in a 4:1 telescope T, to better approximate a plane wave and is then incident on the slit D. The diffraction pattern is observed on the screen S, which is in the focal plane of the lens L. Thus, we observe the image of the pattern formed at infinity. The slit width was $d = 200$ μm and the focal length $f = 50$ cm, so that the first minimum appears at a distance x from the principal maximum

$$x = f\theta \sim f \sin\theta = f(\lambda/d) = 1.68 \text{ mm},$$

where we used $\lambda = 633$ nm. A picture of the diffraction pattern taken with a CCD camera is shown in Fig. 5.4. The central spot saturates the camera.

Instead of using a slit, we can observe the same diffraction pattern by placing a thin wire of width d in the path of the beam. Since it is easier to

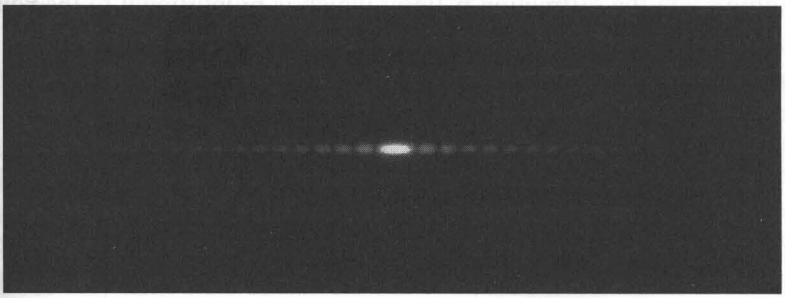


FIGURE 5.4 Diffraction pattern from a thin slit observed in the focal plane obtained with a CCD camera.

obtain thin wires (or hairs) than to manufacture thin slits, the former are often used for demonstrating diffraction. That the two patterns are equivalent (*except* in the forward direction) is known as *Babinet's principle*. To illustrate the principle we assume that the incident plane wave is "uniform and infinite" in extent in the x direction. Thus, the amplitude of the wave is independent of x, $A(x) = A$. Immediately after the *slit*, the amplitude B is given by

$$\begin{aligned} B(x) &= A \qquad -d/2 < x < d/2 \\ B(x) &= 0 \qquad\quad |x| > d/2. \end{aligned} \tag{5.9}$$

In the presence of the obstacle the amplitude C of the wave, just past the obstacle, is given by

$$\begin{aligned} C(x) &= 0 \qquad -d/2 < x < d/2 \\ C(x) &= A \qquad\quad |x| > d/2. \end{aligned} \tag{5.10}$$

Combining Eqs. (5.9) and (5.10) we can write

$$C(x) = A - B(x) \qquad \text{(valid for all } x\text{)}. \tag{5.11}$$

We know that when the amplitude is constant for all x, the wave propagates only in the $\theta = 0$ direction. Thus for angles $\theta \neq 0$, the constant amplitude does not contribute, and Eq. (5.11) becomes

$$C(x) = -B(x) \qquad \text{(valid for } \theta \neq 0\text{)}. \tag{5.12}$$

It follows that the diffraction pattern, which is proportional to the square of the amplitude

$$|C(x)|^2 = |B(x)|^2,$$

is the same in both cases. Equation (5.5) remains valid but with the direction $\theta = 0$ excluded.

Another case of interest arises when instead of a slit a square aperture is used. The result is shown in Fig. 5.5 and consists, primarily, of two single-slit diffraction patterns along the x and y directions. The intensity of the maxima in directions differing from the x or y axes decreases very rapidly.

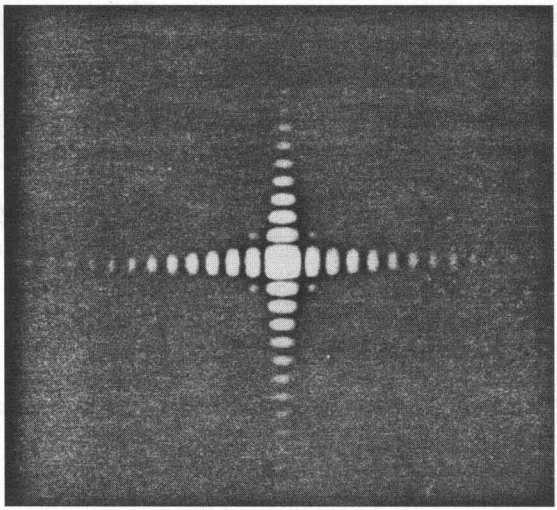


FIGURE 5.5 Diffraction from a square aperture.

5.3. CALCULATION OF THE DIFFRACTION PATTERN

To obtain an expression for the diffraction pattern formed by an aperture, we will make use of the Huygens–Fresnel principle. The principle states that every point on the aperture D is a source of *spherical* wavelets with amplitude and phase determined by the incident wave. These "secondary" wavelets propagate at all angles and interfere at every point in the observation plane to determine the diffracted wave amplitude. We take the incident wavefront parallel to the aperture plane, whereas the observation plane is located at infinity (Fraunhofer diffraction). This is approximated in the sketch of Fig. 5.6 where we show both the aperture and observation planes and two typical rays to the observation point P'. We need be concerned only with the transverse coordinates. In the aperture plane, the point M is specified by the coordinates ζ, η, whereas in the observation plane, the point P' is specified by x', y'.

Because we observe at *infinity*, R, the distance from O to the observation point is very large (and equals OS') as compared to the dimensions of the aperture; therefore, rays OP' and MP' are to be considered as *parallel.* Then the path difference between the ray OP' (from the coordinate origin to P') and the ray MP' (from the source point to P') is the length OB

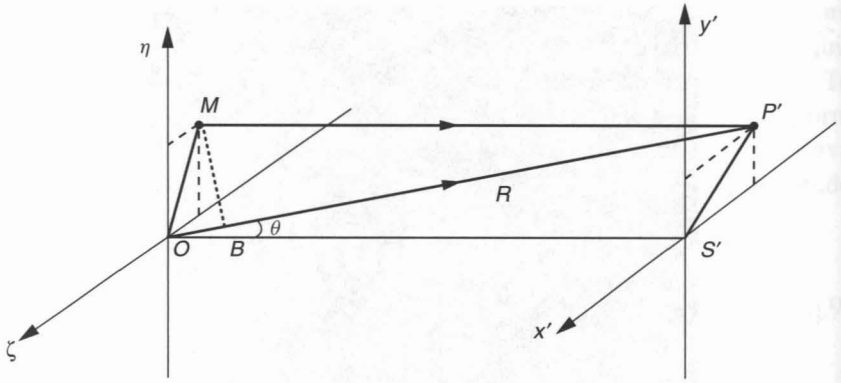


FIGURE 5.6 Coordinate systems in the aperture and observation planes for calculating diffraction.

where MB is perpendicular to OP'. If we designate by $\mathbf{q}$ the unit vector along the ray OP' we obtain for OB

$$OB = \mathbf{OM} \cdot \mathbf{q} = \zeta \left(\frac{x'}{R}\right) + \eta \left(\frac{y'}{R}\right) = [\zeta x' + \eta y']/R. \tag{5.13}$$

The *direction cosines* of the vector OP' are

$$(x'/R) = u \qquad \text{and} \qquad (y'/R) = v$$

and are well defined whether R is finite or tends to infinity. The phase difference between the two rays is

$$\Delta\phi = \frac{2\pi}{\lambda} [\zeta u + \eta v]. \tag{5.14}$$

Without loss of generality we set the reference phase of the ray OP' equal to 0; we can then write for the contribution of the source point M to the amplitude at the observation point P'

$$dA'(x', y') = e^{i\frac{2\pi}{\lambda}[\zeta u + \eta v]} d\zeta d\eta. \tag{5.15}$$

For simplicity we dropped the time-dependence $e^{-i\omega t}$; $d\zeta d\eta$ is the differential element of the aperture at the point M.

To obtain the amplitude at the point P' we must integrate the contribution from all source points. If the amplitude and phase of the incident wave are *constant* over the aperture, we directly integrate Eq. (5.15). For the case of a square aperture with dimensions $2\zeta_0$ and $2\eta_0$ the integral is elementary,

$$A'(x', y') = \int_{-\eta_0}^{\eta_0} \int_{-\zeta_0}^{\zeta_0} e^{i\frac{2\pi}{\lambda}[\zeta u + \eta v]} d\zeta d\eta$$

$$= 4\zeta_0\eta_0 \left[\frac{\sin\left(\frac{2\pi}{\lambda}\zeta_0 u\right)}{\frac{2\pi}{\lambda}\zeta_0 u}\right]\left[\frac{\sin\left(\frac{2\pi}{\lambda}\eta_0 v\right)}{\frac{2\pi}{\lambda}\eta_0 v}\right]. \tag{5.16}$$

The intensity is given by the square of the amplitude

$$I(x', y') = 16\zeta_0^2\eta_0^2 \left[\frac{\sin\left(\frac{2\pi}{\lambda}\zeta_0 u\right)}{\frac{2\pi}{\lambda}\zeta_0 u}\right]^2 \left[\frac{\sin\left(\frac{2\pi}{\lambda}\eta_0 v\right)}{\frac{2\pi}{\lambda}\eta_0 v}\right]^2 \tag{5.17}$$

and is proportional to the *square* of the illuminated (aperture) area. This is typical of diffraction phenomena, as compared to incoherent illumination, which is simply proportional to the area.

In the case of a long *vertical* slit, $\eta_0 \gg \zeta_0$, the intensity vanishes very rapidly for $v \neq 0$. (Note that ηv becomes large and the exponential in Eq. (5.15) oscillates rapidly, its average value tending to zero.) Thus we observe a *horizontal* diffraction pattern confined to the x' axis, as shown in Fig. 5.4. Exactly on the x' axis, $v = 0$ and Eq. (5.17) reduces to

$$I(x', y' = 0) = 16\zeta_0^2\eta_0^2 \left[\frac{\sin\left(\frac{2\pi}{\lambda}\zeta_0 u\right)}{\frac{2\pi}{\lambda}\zeta_0 u}\right]^2 = I_0 \left[\frac{\sin\left(\frac{\pi d}{\lambda}\sin\theta\right)}{\frac{\pi d}{\lambda}\sin\theta}\right]^2. \tag{5.18}$$

In the last step we made use of the relations $u = x'/R = \sin\theta$ (valid for $y' = 0$), where θ is the angle from the z axis and $\zeta_0 = d/2$; we also set $16\zeta_0^2\eta_0^2 = I_0$ to represent the intensity at $\theta = 0$. Note that the above result is exactly that given in Eq. (5.5).

We now consider the case where the amplitude of the incident wave is not constant over the aperture. Such variation can be introduced deliberately by placing a suitable mask over the aperture, or because the incident wave is

modulated in magnitude and/or phase. In this case we express the amplitude on the aperture by

$$F(\zeta, \eta),$$

and the amplitude at the observation point P' (Eq. (5.16)) takes the form

$$A'(x', y') = \int\int F(\zeta, \eta) e^{i\frac{2\pi}{\lambda}(\zeta u + \eta v)} d\zeta d\eta. \tag{5.19}$$

The integration in Eq. (5.19) is over the aperture; however, since the amplitude of the transmitted wave vanishes beyond the aperture boundary, we can extend the integration limits to infinity. With this modification, we see that the far-field amplitude, i.e., in the focal plane of the lens, is the *Fourier transform* of the amplitude in the near field. To explain this statement note that Eq. (5.19) is very similar to the more familiar Fourier transform between the frequency and time domains. If $F(t)$ describes the time dependence of a pulse, then $A(\omega)$ describes the spectrum of the frequencies contained in the pulse

$$A(\omega) = \int_{-\infty}^{+\infty} F(t) e^{-i\omega t}\, dt. \tag{5.20}$$

Similarly, in Eq. (5.19) $F(\zeta, \eta)$ describes the spatial dependence in the aperture plane and $A'(x', y')$ can be thought of as describing the spectrum of "spatial frequencies" $2\pi u/\lambda$ and $2\pi v/\lambda$. We will make use of these concepts in Section 5.4.

5.4. DIFFRACTION FROM A CIRCULAR APERTURE

Instead of a slit we shall now use a circular aperture. Some skill is required in manufacturing such small apertures, but they can also be purchased commercially. The smaller the diameter, the larger the diffraction angle, but the transmitted intensity decreases at the fourth power of the diameter, making observation of the pattern correspondingly more difficult. In the present experiment the aperture diameter was $d = 150$ μm, which is a good compromise for the HeNe wavelength.

To obtain the diffraction pattern, we integrate Eq. (5.15) over the circular aperture. To do so we rewrite Eq. (5.15) in terms of polar coordinates as

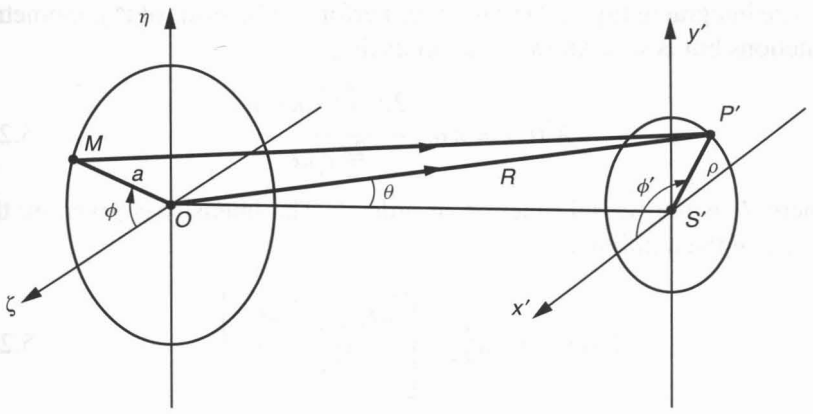


FIGURE 5.7 Coordinate systems in polar coordinates for calculating diffraction.

shown in Fig. 5.7. In the source plane we use the coordinates a, ϕ so that

$$\zeta = a\cos\phi \qquad \eta = a\sin\phi, \tag{5.21}$$

whereas in the observation plane we use ρ, ϕ' so that

$$u = \frac{x'}{R} = \frac{\rho\cos\phi'}{R} = \alpha\cos\phi', \qquad v = \frac{y'}{R} = \frac{\rho\sin\phi'}{R} = \alpha\sin\phi', \tag{5.22}$$

where $\alpha = \rho/R$ is the sine of the radial diffraction angle. Expressed in terms of these new coordinates, the argument of the exponential in Eq. (5.15) becomes

$$i\,\frac{2\pi}{\lambda}\,[\zeta u + \eta v] = i\,\frac{2\pi}{\lambda}\,a\alpha\cos(\phi - \phi'). \tag{5.23}$$

Since the origin of the angles is arbitrary we can set $\phi' = 0$ (the pattern must be rotationally symmetric about the axis). Thus the amplitude at the angle θ, where $\sin\theta = \alpha$, is given by

$$A'(\alpha) = \int_0^{a_0}\int_0^{2\pi} e^{i\frac{2\pi}{\lambda}a\alpha\cos\phi}\,a\,da\,d\phi. \tag{5.24}$$

Here a_0 is the radius of the circular aperture, and we have assumed uniform illumination.

The integral in Eq. (5.24) cannot be performed in terms of trigonometric functions but is well known. One finds that

$$A'(\alpha) = \pi a_0^2 \frac{2J_1\left(\frac{2\pi}{\lambda} a_0 \alpha\right)}{\frac{2\pi}{\lambda} a_0 \alpha}, \tag{5.25}$$

where J_1 is the Bessel function of order 1. The intensity is given by the square of the amplitude

$$I(\alpha) = \left(\pi a_0^2\right)^2 \left[\frac{2J_1\left(\frac{2\pi}{\lambda} a_0 \alpha\right)}{\frac{2\pi}{\lambda} a_0 \alpha}\right]^2. \tag{5.26}$$

We recognize that the intensity is proportional to the *square* of the illuminated area. Since $\alpha = \sin\theta$ is the diffraction angle, Eq. (5.26) is similar to Eq. (5.5) with the replacement of the sine by the J_1 Bessel function.

Equation (5.26) is plotted as a function of its argument, $x = (2\pi/\lambda)a_0\alpha$, in Fig. 5.8. The zeros occur at the following values of x,

$$x_1 = 3.83, \qquad x_2 = 7.02, \qquad x_3 = 10.17, \text{ etc.},$$

whereas the maxima fall in between. The pattern is that of an intense central disk surrounded by alternating dark and bright rings, as shown in Fig. 5.9. The first dark ring occurs at an angle

$$\theta_1 \simeq \sin\theta_1 = \frac{3.83}{\pi} \frac{\lambda}{2a_0} = 1.22 \frac{\lambda}{D}, \tag{5.27}$$

where D is the diameter of the aperture. If the lens used has a focal length f, the radius of the first dark ring on the screen occurs at

$$\rho_1 = 1.22 \left(\frac{f}{D}\right) \lambda. \tag{5.28}$$

Equation (5.27), first obtained by Airy, gives the smallest radius that can be obtained by focusing a beam of wavelength λ with optics specified by the *f-number* (f/D). The shorter the focal length, and the larger the aperture, the smaller the focal spot and thus the resolution of the instrument. The central disk contains 76% of the total intensity.

The experimental setup is the same as shown in Fig. 5.3, except that the slit is replaced by the circular "pinhole." Figure 5.9 is a CCD picture obtained with a 150-μm diameter pinhole. Three dark rings could be

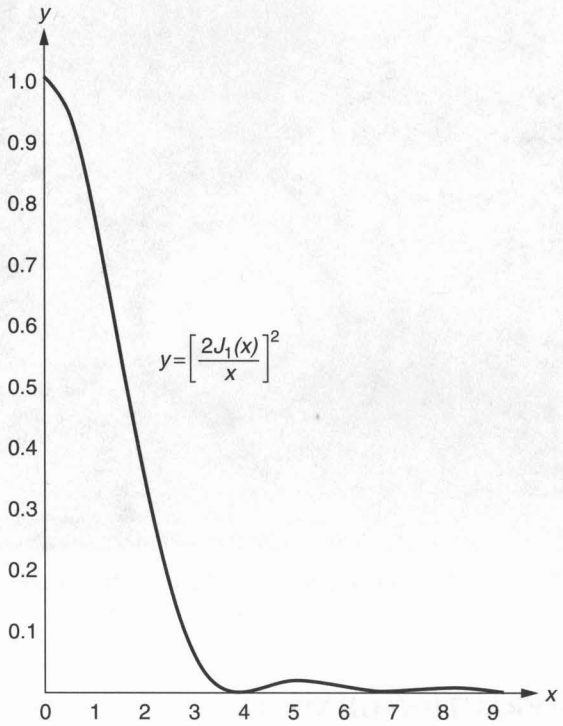


FIGURE 5.8 The intensity distribution for Fraunhofer diffraction from a circular aperture as a function of $x = (2\pi/\lambda)a_0 \sin\theta$; θ is the diffraction angle and a_0 the aperture radius.

measured at the angles

$$\alpha_1 = (5.25 \pm 1) \times 10^{-3} \text{ radians}$$

$$\alpha_2 = (10.5 \pm 2) \times 10^{-3} \text{ radians}$$

$$\alpha_3 = (14.5 \pm 2) \times 10^{-3} \text{ radians}.$$

Using the values for the zeros of J_1 as given previously, we obtain the corresponding values for λ/D

$$(\lambda/D) = 4.3 \times 10^{-3},\ 4.7 \times 10^{-3},\ 4.5 \times 10^{-3}.$$

These results are self-consistent and predict a pinhole diameter $D = 140\ \mu\text{m}$, in good agreement with the "nominal" value.

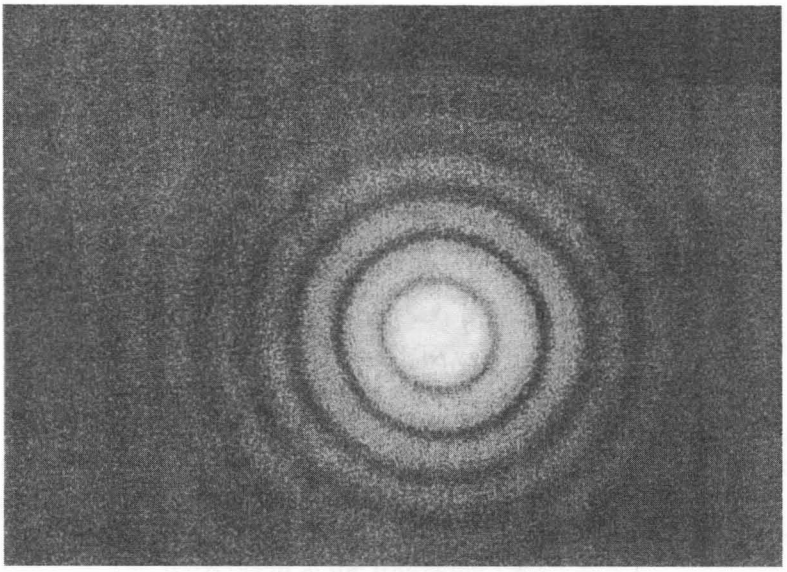


FIGURE 5.9 Observed diffraction pattern of a HeNe beam from a pinhole of diameter 150 μm.

5.5. THE DIFFRACTION GRATING

We have already made use of the diffraction grating and discussed the physical principles in Chapter 1. Here we will carry out a more detailed analysis and demonstrate a compact spectrometer using a digital readout.

If instead of a single slit, two slits are illuminated by a plane wavefront, a series of interference fringes parallel to the slits will appear on a far screen. This is the classical experiment of Thomas Young (1800) shown in Fig. 5.10a. If the spacing between the slits is d, the intensity distribution on the screen is[1]

$$I(\theta) = 4I_0 \cos^2\left(\frac{\pi d}{\lambda}\sin\theta\right). \tag{5.29}$$

The angle θ is measured, as usual, with respect to the normal to the plane containing the slits. If one of the slits is blocked, the fringes disappear and the transmitted intensity is I_0.

[1] We take the wavefront parallel to the plane in which lie the slits.

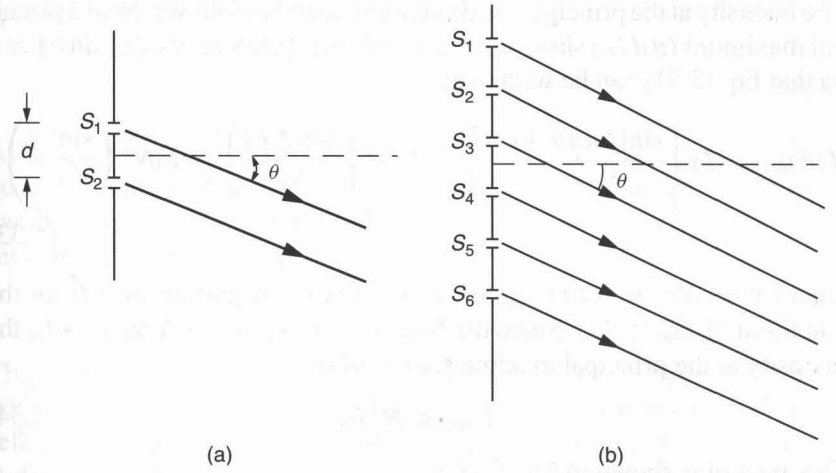


FIGURE 5.10 (a) Young's two-slit experiment. (b) Multiple-slit interference, the diffraction grating.

In Eq. (5.29) we have not included the effects of diffraction due to the width of the slits. Let the slit width be a. Then Eq. (5.29) is modulated by the diffraction pattern of Eq. (5.5), and we obtain for the intensity distribution

$$I(\theta) = 4I_0 \cos^2\left(\frac{\pi d}{\lambda}\sin\theta\right)\left[\frac{\sin\left(\frac{\pi a}{\lambda}\sin\theta\right)}{\frac{\pi a}{\lambda}\sin\theta}\right]^2. \tag{5.30}$$

If instead of two slits, several equidistantly spaced slits are illuminated by the wavefront, the interference maxima become much sharper, and the interference pattern is given by

$$I(\theta) = I_0\left[\frac{\sin\left(N\frac{\pi d}{\lambda}\sin\theta\right)}{\sin\left(\frac{\pi d}{\lambda}\sin\theta\right)}\right]^2. \tag{5.31}$$

Here d is the spacing between the slits, and N the total number of slits; we have not included the effects of diffraction because in practical applications the slits are so narrow that the modulation is not important. Note that Eq. (5.31) reduces to Eq. (5.29) for $N = 2$, as it must.

What is of particular interest is that the pattern contains *principal* maxima when the denominator of Eq. (5.31) becomes zero, namely when

$$\sin\theta = \pm n\lambda/d, \qquad n = 0, 1, 2, \ldots. \tag{5.32}$$

The intensity at the principal maxima can be found as follows. Near a principal maximum $(\pi d/\lambda)\sin\theta = n\pi + \epsilon$ and therefore $\sin[(\pi d/\lambda)\sin\theta] \simeq \epsilon$ so that Eq. (5.31) can be written as

$$I(\theta)_{\max} = I_0\left[\frac{\sin[N(n\pi+\epsilon)]}{\epsilon}\right]^2 = I_0N^2\left[\frac{\sin(N\epsilon)}{N\epsilon}\right]^2 = I_0N^2\left(\frac{\sin^2 x}{x}\right), \tag{5.33}$$

where $x = N\epsilon = N\pi(d/\lambda)\Delta\theta$, and $\Delta\theta$ is the departure of θ from the condition of Eq. (5.32). Since the function $(\sin x/x)^2 \to 1$ as $x \to 0$, the intensity at the principal maxima $[\Delta\theta = 0]$ is

$$I_{\max} = N^2 I_0. \tag{5.34}$$

This pattern is shown in Fig. 5.11.

The width of the principal maxima is given by the first minimum of the function $(\sin x/x)$, which occurs when $x = \pm\pi$, namely

$$\Delta\theta = \pm\frac{\lambda}{Nd}. \tag{5.35}$$

Note that (Nd) is the total extent of the region covered by the slits. Thus the principal maxima are as narrow as if the wavefront diffracted from a slit of width Nd. By combining Eq. (5.35) with Eq. (5.32) we can express

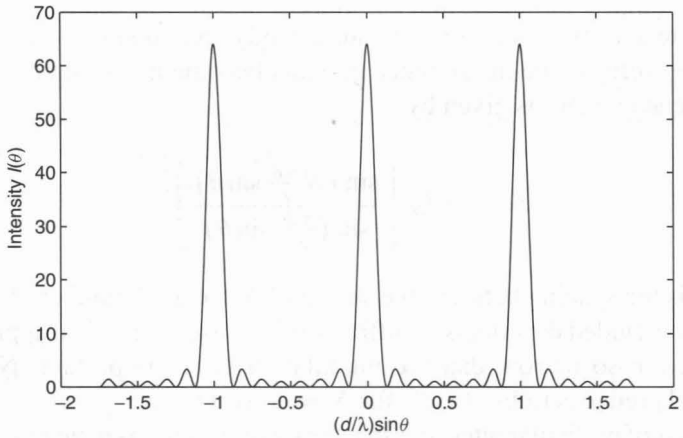


FIGURE 5.11 Different orders of monochromatic light scattered from a grating. Note that the principal maxima are very narrow peaks, whereas the secondary maxima are suppressed. Plotted for $N = 5$.

the resolution of the system of N slits by

$$\frac{\Delta\lambda}{\lambda} = \frac{1}{Nn}\cos\theta \simeq \frac{1}{Nn}. \tag{5.36}$$

A *diffraction grating* is equivalent to such a system of many slits and can be used either in transmission or in reflection. The angle of incidence θ_i can be different from the normal to the grating, in which case Eq. (5.32) must be modified to read

$$\sin\theta_i - \sin\theta_r = \pm n\frac{\lambda}{d}, \qquad n = 0, 1, 2, \ldots. \tag{5.37}$$

The diffraction angle is θ_r and is taken positive if it is *opposite* from θ_i with respect to the normal. These definitions are shown in Fig. 5.12. For a reflection grating $n = 0$ corresponds to specular reflection ($\sin\theta_r = \sin\theta_i$). Reflection gratings are often manufactured so as to enhance reflection at particular angles. Recall that Eq. (5.37) was already used in Chapter 1 (see Eq. (1.16)).

The arrangement used in the laboratory is shown in Fig. 5.13. The light source is focused on the slit and the emerging beam is made parallel by lens L1, which has focal length $f_1 = 20$ cm. The parallel beam is incident on the 4×4 cm^2 grating, which has 1200 lines/mm. The angle of incidence was chosen to be $\theta_i = 55.7°$. The beam diffracted in first order was focused with lens L2, identical to L1, onto the "reticon" where it formed an image of the slit.

The reticon is a linear array of pixels, which can be read out on an oscilloscope. In the present case the array contained 128 pixels; the clock speed was 80 kHz so that a pixel is read out every $\Delta t = 12.5$ μs. The pixel

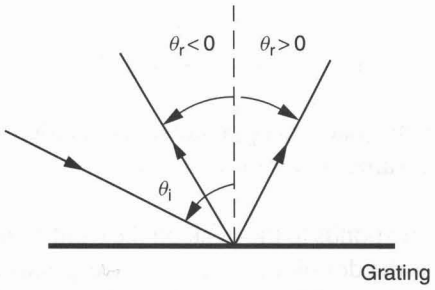


FIGURE 5.12 The convention used for labeling the incidence and reflection angles for a reflection grating.

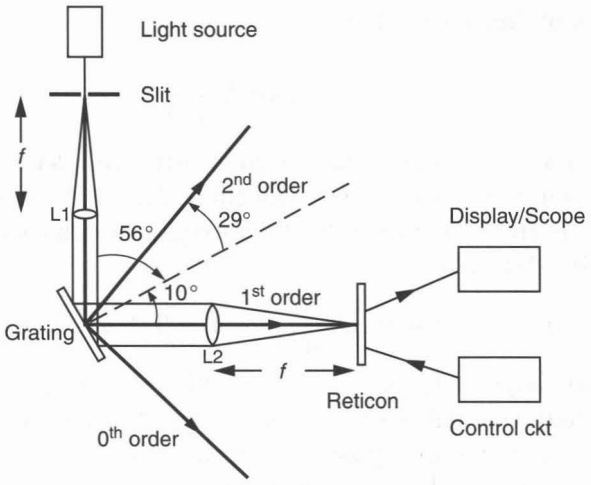


FIGURE 5.13 Layout of a simple grating spectrometer read out by a reticon (a one-dimensional solid-state detector array).

size was $\Delta x_0 = 100\ \mu$m for a total array length of 1.28 cm. Thus we have the conversion factor

$$\Delta x = 8\ \mu\text{m}/\mu\text{s}, \tag{5.38}$$

and since $\Delta x = f\Delta\theta$, $f = 20$ cm

$$\Delta\theta = 0.04\ \text{mrad}/\mu\text{s}. \tag{5.39}$$

The spectrum of a Hg arc lamp is shown in Fig. 5.14a. The horizontal scale (sweep speed) corresponds to 200 μs/cm. The spectrum was observed in first order, and from Eq. (5.37) with $\theta_i = 55.7°$, $d = 10^{-3}/(1.2 \times 10^3)$ m we find that for the green line of Hg ($\lambda_g = 546.1$ nm)

$$\sin\theta_r = \sin\theta_i - \frac{\lambda}{d} = 0.170,$$

namely $\theta_r = 9.8°$ in the quadrant opposite to the incident beam. The second order appears in the same quadrant as the incident beam at $\theta_r = 29°$ as shown in Fig. 5.13.

The green line corresponds to the peak on the right-hand side of the graph (Fig. 5.14a), whereas the doublet on the left corresponds to the yellow lines ($\lambda_1 = 577.7$ nm and $\lambda_2 = 579.1$ nm). Knowledge of these wavelengths allows us to make a more precise calibration of the spectrometer, including

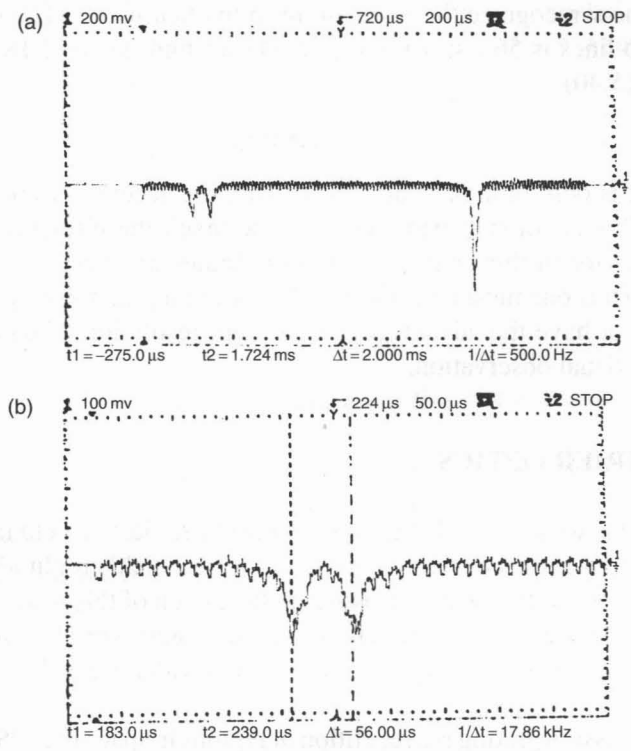


FIGURE 5.14 The observed spectrum (a) of the green line and yellow doublet of the Hg spectrum obtained with the spectrometer of Fig. 5.13b. (b) The yellow doublet on an expanded scale.

misalignment and other instrumental effects. Differentiating Eq. (5.37) with θ_i fixed, we obtain

$$n\Delta\lambda/d = -\cos\theta_r \Delta\theta_r. \tag{5.40}$$

In our case $n = 1$, $\cos\theta_r = 0.99$ and $\Delta\lambda$, from the first yellow line λ_2 to the green line λ_g, is $\Delta\lambda = 33$ nm, or $\Delta\theta = 40.0$ mrad. The time interval between these lines as measured off Fig. 5.14a is $\Delta t = 1030$ µs, and thus the calibration

$$\Delta\theta = 39 \times 10^{-3} \text{ mrad/µs} \tag{5.41}$$

in close agreement with the direct calculation.

To measure the fine structure of the yellow doublet the sweep speed is increased so that the scale factor is 50 µs/cm as shown in Fig. 5.14b.

One can now recognize the response of individual pixels. The separation of the two lines is 56 μs; using Eq. (5.41) we find $\Delta\theta = 2.18$ mrad and from Eq. (5.40)

$$\Delta\lambda = 1.8 \text{ nm}.$$

Our result is only in modest agreement with the accepted value of $\Delta\lambda = 1.4$ nm. This is not surprising because one pixel, the ultimate resolution of our detector in this configuration, contributes an uncertainty of $\delta\lambda = 0.42$ nm. Thus one must be cautious when using digital techniques, which often do not have the advantages of the high resolution of photographic film or of visual observation.

5.6. FOURIER OPTICS

In Eq. (5.19), we showed that the amplitude of the electric field in the focal plane of a lens is the Fourier transform of the near-field amplitude incident on the lens. We will now give a physical discussion of this result and show how it can be used in practice. These considerations were first introduced by E. Abbe in Jena, Germany, but found much wider use as lasers became available.

A transmission grating is a repetition of regions in space that alternatively transmit/absorb the incident wavefront; we can represent the transmission of the grating by the "square-wave" function shown in Fig. 5.15a. We are immediately reminded of the analogous square-wave function of time that has period T, and thus frequency $\nu = 1/T$. Therefore we can assign to the

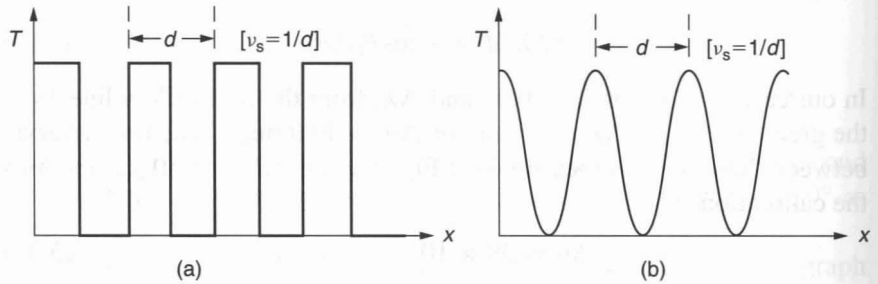


FIGURE 5.15 (a) Representation of the transmission of a grating; the spatial spectrum contains the fundamental frequency $\nu_S = 1/d$ and its higher harmonics. (b) If the transmission is sinusoidal, only the frequency $\nu_S = 1/d$ is present in the scattered wave.